CS257 Linear and Convex Optimization

Homework 1

Due: September 21

September 14, 2020

1. Let $f(\boldsymbol{x}) = f(x_1, x_2, x_3) = x_1^2 + e^{x_2} + e^{-x_2} + 2x_3^4$.

(a). Does f have a minimum and a maximum over the set $D = \{\boldsymbol{x} : x_1^2 + 2x_2^2 + 3x_3^2 \leq 1\}$?

(b). Does f have a minimum and a maximum on $\mathbb{R}^3$?

You don't have to solve the optimization problems!

Solution. (a). f is continuous and D is compact, so f has a minimum and maximum over D by the Extreme Value Theorem.

(b). f is coercive, so f has a minimum, but no maximum. The minimum is $(0, 0, 0)$.

□

2. Let $f(\boldsymbol{x}) = x_1^2 + x_1x_2 + 2x_2^2 - 3x_1 - 5x_2$.

(a). Is $f(x)$ coercive? Hint: use $x_1x_2 \geq -(x_1^2 + x_2^2)/2$.

(b). Find the minimum and maximum of $f(\boldsymbol{x})$ over $\mathbb{R}^2$ if they exist.

Solution. (a). Note

$$f(\boldsymbol{x}) \geq x_1^2 - \frac{1}{2}(x_1^2 + x_2^2) + 2x_2^2 - 3x_1 - 5x_2 = \left(\frac{1}{2}x_1^2 - 3x_1\right) + \left(\frac{3}{2}x_2^2 - 5x_2\right)$$

Clearly the RHS goes to $+\infty$ as $\|\boldsymbol{x}\| \to \infty$, so $f(\boldsymbol{x})$ is coercive.

(b). Since f is coercive, it has no maximum. For the minimum, by the first-order necessary condition,

$$\nabla f(\boldsymbol{x}) = 2\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{b} = (2x_1 + x_2 - 3, x_1 + 4x_2 - 5)^T = 0 \implies \boldsymbol{x}^* = (1, 1)^T$$

The Hessian matrix is

$$\nabla^2 f(\boldsymbol{x}) = 2\boldsymbol{Q} = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$$

It is positive definite, since $\det(2) = 2 > 0$, $\det \nabla^2 f = 7 > 0$. Thus $\boldsymbol{x}^* = (1, 1)^T$ is a local minimum. Since f is coercive, it has a global minimum. Since it has only one local minimum, it must be the global minimum.

□

3. For what value of α is the following matric positive definite?

$$A = \begin{pmatrix} 1 & \alpha & -1 \\ \alpha & 4 & 2 \\ -1 & 2 & 4 \end{pmatrix}$$

Solution.

$$\det(1) > 0, \quad \det\begin{pmatrix} 1 & \alpha \\ \alpha & 4 \end{pmatrix} = 4 - \alpha^2 > 0, \quad \det\begin{pmatrix} 1 & \alpha & -1 \\ \alpha & 4 & 2 \\ -1 & 2 & 4 \end{pmatrix} = -4\alpha^2 - 4\alpha + 8 > 0$$

$$\alpha \in (-2, 1)$$

□

4. Are the following matrices positive definite, positive semidefinite, negative definite, negative semidefinite, or indefinite?

(a). Solve the following by writing out the eigenvalues. You can use **numpy.linalg.eigvals** to do it.

$$A = \begin{pmatrix} 7 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{pmatrix},$$

(b). Solve the following by writing out the eigenvalues. Please calculate by hand.

$$B = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

(c). Solve the following two by checking the principal minors.

$$C = \begin{pmatrix} -5 & 2 & 2 \\ 2 & -6 & 0 \\ 2 & 0 & -4 \end{pmatrix}, \quad D = \begin{pmatrix} -1 & 1 & -2 \\ 1 & -1 & 1 \\ -2 & 1 & 1 \end{pmatrix}$$

Solution. (a). We can obtain the eigenvalues for A:

$$8.04096459, 3.87053872, 2.08849669.$$

Therefore, A is positive definite.

(b). We can obtain the eigenvalues for B:

$$4, 2, 3.$$

Therefore, B is positive definite.

(c). Clearly C is not positive (semi)definite as $\det(-5) < 0$. Check the leading principal minors of $-C$,

$$\det(5) = 5 > 0, \quad \det\begin{pmatrix} 5 & -2 \\ -2 & 6 \end{pmatrix} = 26 > 0, \det -C = 80 > 0.$$

C is negative definite.

Clearly D is not positive (semi)definite as $\det(-1) < 0$. Check the leading principal minors of $-D$,

$$\det(1) = 1 > 0, \quad \det\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = 0, \quad \det -D = -1 < 0$$

D is indefinite.

□